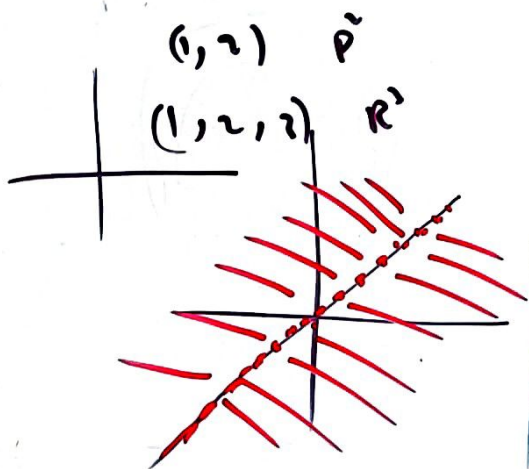


$$\text{if } z = f(x, y) = x^2 + 2y^3$$

$$f(3, -1) = \dots$$

$$f(3, -1) = (3)^2 + 2(-1)^3$$

$$= 9 - 2 = 7$$



functions of two variables

Domain of functions

find the domain of functions

$$\textcircled{1} z = 2x^2 + 5xy^2$$

the function is defined and real points of xy-plane

$$D = \{(x, y) : x, y \in \mathbb{R}\}$$

$$\textcircled{2} z = \frac{x^2 + y^2 - 2xy}{x - 2y}$$

$$x - 2y = 0$$

$$x = 2y$$

the function is defined all points in the xy-plane

except also $x = 2y$

$$D = \{(x, y) \in \mathbb{R}^2 : x \neq 2y\}$$

$$\textcircled{3} z = \ln(1 - x^2 - y^2)$$

$$1 - x^2 - y^2 > 0$$

$$-x^2 - y^2 > -1$$

$$x^2 + y^2 < 1$$

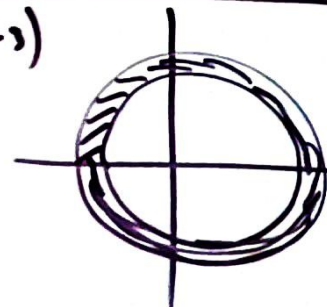


$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$

$$f(x, y) = \sin^{-1}(x^2 + y^2 - 3)$$

$$-1 \leq x^2 + y^2 - 3 \leq 1$$

$$2 \leq x^2 + y^2 \leq 4$$



Continuity الاتصال

$f(x,y)$ is said to be continuous
at the point (a,b)

if ① $f(a,b)$ is defined

② $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x,y)$ exists

③ $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x,y) = f(a,b)$

functions of two variables

show that the function

$$f(x,y) = \begin{cases} 3xy & (x,y) \neq (1,2) \\ 0 & (x,y) = (1,2) \end{cases}$$

is dis continuous at $(1,2)$

sol

① $f(1,2) = 0$

② $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} f(x,y) = \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} 3xy = 3(1)(2) = 6$

$$f(1,2) \neq \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} f(x,y)$$

dis continuous at $(1,2)$

Continuity الاتصال

In vestigate the continuity of
the function

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} \\ 0 \end{cases}$$

$$(x,y) \neq (0,0)$$

$$(x,y) = (0,0)$$

sol

$$1) f(0,0) = 0 \rightarrow (1)$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2+y^2}$$

functions of two variables

$$y = mx$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2+y^2} = \lim_{x \rightarrow 0} \frac{mx^2}{x^2+m^2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{mx^2}{x^2(1+m^2)}$$

$$= \frac{m}{1+m^2}$$

limit does not exist

$\therefore f(x,y)$ is not continuous at $(0,0)$

lim et

$$\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} \frac{2x^2y}{x^2+y^2+1} = \frac{2(1)^2(2)}{(1)^2+(2)^2+1} \\ = \frac{4}{1+4+1} = \frac{4}{6} = \frac{2}{3}$$



functions of two variables

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x-y}{x+y} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} \frac{x-y}{x+y} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{x-0}{x+0} \right) = \lim_{x \rightarrow 0} \frac{x}{x} = 1 \rightarrow (1)$$

$$\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} \frac{x-y}{x+y} \right) = \lim_{y \rightarrow 0} \left(\frac{0-y}{0+y} \right)$$

$$= \lim_{y \rightarrow 0} \frac{-y}{y} = -1 \rightarrow (2)$$

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x,y) \right) \neq \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x,y) \right)$$

does not exist

let $y = mx$ $\begin{matrix} mx \rightarrow 0 \\ x \rightarrow 0 \end{matrix}$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x-y}{x+y} = \lim_{x \rightarrow 0} \left(\frac{x-mx}{x+mx} \right)$$

$$= \lim_{x \rightarrow 0} \frac{x(1-m)}{x(1+m)} = \frac{1-m}{1+m}$$

does not exist

functions of two variables

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2 + y^2}$$

$$\frac{0}{0}$$

sol

$$y = mx$$

$$\lim_{x \rightarrow 0} \frac{x(mx)}{x^2 + (mx)^2}$$

$$= \lim_{x \rightarrow 0} \frac{mx^2}{x^2 + m^2 x^2}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 m}{x^2 (1 + m^2)}$$

$$= \frac{m}{1 + m^2}$$

limits do not exist

Partial derivatives

$$z = f(x, y)$$

$$\textcircled{1} \frac{\partial z}{\partial x} \quad y \text{ ثابت}$$

$$\textcircled{2} \frac{\partial z}{\partial y} \quad x \text{ ثابت}$$

$$\textcircled{3} \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right)$$

$$\textcircled{4} \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right)$$

$$\textcircled{5} \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)$$

$$\textcircled{6} \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Functions of two variables

$$\frac{\partial}{\partial x} (a) = 0$$

$$\frac{\partial}{\partial x} (a f(x)) = a \frac{\partial f(x)}{\partial x}$$

$$\frac{\partial}{\partial x} \left(\frac{f(x)}{a} \right) = \frac{1}{a} \frac{\partial f(x)}{\partial x}$$

$$\frac{\partial}{\partial x} \left(\frac{x}{a} \right) = \frac{1}{a}$$

$$\frac{\partial}{\partial x} \left(\frac{a}{x^n} \right) = \frac{-a n}{x^{n+1}}$$

$$\frac{\partial}{\partial x} \left(\frac{3}{x} \right) = \frac{-3}{x^2}$$

$$\frac{\partial}{\partial x} (\tan^{-1}(x)) = \frac{(x)'}{1+x^2} = \frac{1}{1+x^2}$$

Partial derivatives

$$z = f(x, y) = x^3 + y^3 - 3axy$$

$$(1) \frac{\partial z}{\partial x}$$

$$\frac{\partial z}{\partial x} = 3x^2 + 0 - 3ay \quad (1)$$

$$\frac{\partial z}{\partial x} = 3x^2 - 3ay$$

(y) ثابت

$$(2) \frac{\partial z}{\partial y} = 0 + 3y^2 - 3ax \quad (1)$$
$$= 3y^2 - 3ax$$

$$(3) \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)$$
$$= \frac{\partial}{\partial x} (3y^2 - 3ax)$$
$$= 0 - 3a = -3a$$

Functions of two variables

$$(3) \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right)$$

$$= \frac{\partial}{\partial x} (3x^2 - 3ay)$$

$$= 6x - 0$$

$$= 6x$$

$$(6) \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)$$
$$= \frac{\partial}{\partial y} (3x^2 - 3ay)$$
$$= 0 - 3a = -3a$$

$$(4) \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right)$$

$$= \frac{\partial}{\partial y} (3y^2 - 3ax)$$

$$= 6y$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

✓

partial derivatives

If $u = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$

show that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

sol

$$\frac{\partial u}{\partial x} = (2x) \tan^{-1}\left(\frac{y}{x}\right) + \left(\frac{-y}{1 + \frac{y^2}{x^2}}\right)x^2 - \left[y^2 \frac{\frac{1}{y}}{1 + \frac{x^2}{y^2}}\right]$$

$$= 2x \tan^{-1}\left(\frac{y}{x}\right) - \frac{y x^2}{x^2 + y^2} - \frac{y^3}{y^2 + x^2}$$

$$= 2x \tan^{-1}\left(\frac{y}{x}\right) + \frac{-y x^2 - y^3}{x^2 + y^2}$$

$$= 2x \tan^{-1}\left(\frac{y}{x}\right) - y \frac{(x^2 + y^2)}{x^2 + y^2} = 2x \tan^{-1}\left(\frac{y}{x}\right) - y$$

functions of two variables

$$\frac{\partial u}{\partial y} = x^2 \cdot \frac{\frac{1}{x}}{1 + \frac{y^2}{x^2}} - \left[2y \tan^{-1}\left(\frac{x}{y}\right) + y^2 \left(\frac{-\frac{x}{y^2}}{1 + \frac{x^2}{y^2}}\right)\right]$$

$$= \frac{x^3}{x^2 + y^2} - 2y \tan^{-1}\left(\frac{x}{y}\right) + \frac{x y^2}{y^2 + x^2}$$

$$= \frac{x^3 + x y^2}{x^2 + y^2} - 2 \tan^{-1}\left(\frac{x}{y}\right)$$

$$= \frac{x(x^2 + y^2)}{x^2 + y^2} - 2 \tan^{-1}\left(\frac{x}{y}\right) = x - 2 \tan^{-1}\left(\frac{x}{y}\right)$$

$$\frac{\partial u}{\partial y} = x - 2 \tan^{-1}\left(\frac{x}{y}\right)$$

Partial derivatives

If $u = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$ functions of two variables

show that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

sol

$$\frac{\partial u}{\partial y} = x^2 \frac{1}{1 + \frac{y^2}{x^2}} - \left[2y \tan^{-1}\left(\frac{x}{y}\right) + y^2 \frac{-\frac{x}{y^2}}{1 + \frac{x^2}{y^2}} \right]$$

$$= \frac{x^3}{x^2 + y^2} - 2y \tan^{-1}\left(\frac{x}{y}\right) + \frac{xy}{y^2 + x^2}$$

$$= \frac{x(x^2 + y^2)}{y^2 + x^2} - 2y \tan^{-1}\left(\frac{x}{y}\right) = x - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

مساوی

$$\therefore \frac{\partial u}{\partial y} = x - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} \left(x - 2y \tan^{-1}\left(\frac{x}{y}\right) \right)$$

$$= 1 - 2y \frac{1}{1 + \frac{x^2}{y^2}} = 1 - \frac{2y^2}{y^2 + x^2} = \frac{y^2 + x^2 - 2y^2}{y^2 + x^2} = \frac{x^2 - y^2}{y^2 + x^2}$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial y} \left(2x \tan^{-1}\left(\frac{y}{x}\right) - y \right)$$

$$= 2x \frac{1}{1 + \frac{y^2}{x^2}} - 1 = \frac{2x^2}{x^2 + y^2} - 1 = \frac{2x^2 - x^2 - y^2}{x^2 + y^2} = \frac{x^2 - y^2}{x^2 + y^2} \rightarrow (2)$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

partial derivatives

$$y = \log(x^2 + y^2)$$

$$\frac{\partial z}{\partial x}$$

$$\frac{\partial z}{\partial y}$$

sol

$$(1) \frac{\partial z}{\partial x} = \frac{2x + 0}{x^2 + y^2} = \frac{2x}{x^2 + y^2}$$

$$(2) \frac{\partial z}{\partial y} = \frac{0 + 2y}{x^2 + y^2} = \frac{2y}{x^2 + y^2}$$

functions of two variables

$$z = x^2 - x \sin\left(\frac{x}{y}\right)$$

$$\frac{\partial z}{\partial x}$$

$$\frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial x} = 2x - \left[(1) \sin\left(\frac{x}{y}\right) + x \cos\left(\frac{x}{y}\right) \cdot \left(-\frac{1}{y}\right) \right]$$

$$= 2x - \sin\left(\frac{x}{y}\right) - \frac{x}{y} \cos\left(\frac{x}{y}\right)$$

$$\frac{\partial z}{\partial y} = 0 - x \cos\left(\frac{x}{y}\right) \cdot \left(-\frac{x}{y^2}\right)$$

$$= \frac{x^2}{y^2} \cos\left(\frac{x}{y}\right)$$

✓